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Mechanisms of the Digital Transformation; A Cause for Policy Reform

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In all my experience working in the field of economics (admittedly not within the same context as this article), there have been two things that have fascinated me the most. The first – which I will not discuss any further – is trade, and the second is a phenomenon known as the digital transformation. It was only once realizing the magnitude of its transformative effects on the global economy that I began to take an interest. Its ramifications are ubiquitous throughout society and cause for widespread policy reform.

The digital transformation is also commonly referred to as digitalization. Although these two terms are interchangeable, they must not be confused with digitization, which refers to the actual encoding of information into binary (bits and bytes, etc.). Although digitization is very clear cut and not conceptually challenging, there lies disparity among professionals in the term digitalization (or the digital transformation). It can be thought of as "the changes produced by different forms of digitization, the resulting applications, systems, platforms, and the effects on economic and social activity" (Ahmad & Ribarsky, 2018). The concept of digitalization is much more widespread and permeates throughout the entire economy.

To produce an estimate of the digital economy is a difficult task, as it is not always clear whether some economic activity should be included in the estimate (Ahmad & Ribarsky, 2018). Regardless of its size, the digital economy is irrefutably an important element of the global economy and in recent years has begun undermining policies established in a pre-digital era.

An overwhelming amount of policies in today's society are based on concepts such as tangible products and assets, fixed geographic boundaries, supply and demand scenarios in which scarcity is a limitation, and transaction costs that limit the scale and scope of economic activity. The digital transformation is crippling these policies, causing inefficiencies that must be addressed (OECD, 2019).

With such drastic changes to our economy underway, an opportunity is presented to greatly increase policy contributions towards various societal objectives. By aligning both old and new policies with the digital economy, improvements to the welfare and more extensive growth can be achieved (OECD, 2019). This, however, requires an understanding of the mechanisms through which digitalization is shaping the global economic landscape. Although understanding digitalization can be a particularly daunting task in such a rapidly evolving economy, in 2019 the OECD published a paper addressing the key aspects of digitalization which hold significance in the designing of effective 21st-century policies. The paper outlines several "vectors" through which digitalization impacts almost all domains of policy.

The vectors can be organized into three major groups:-

1. Scale, scope, and speed

2. Ownership, assets, and economic value

3. Relationships, markets, and ecosystems

Each of these groups reflects an area in which digitalization affects economic activity. They are not exclusive, and each group is heavily associated with the others. To begin, the digitization of information has increased the scale, scope, and speed of economic activity remarkably. Not only that, but the cost of exploiting digital devices in recent decades has also become exponentially cheaper, affecting the nature of assets that generate value, how ownership is imparted, and where value is generated. These changes have altered the structure of markets which are now supported by all kinds of digital products, creating entirely new exchange environments (OECD, 2019).

Examining the vectors proposed by the OECD in detail, we obtain a more refined view of how digitalization is transforming the economy. The following is a summary of the mechanisms through which each of the three groups (and their respective "vectors") have been able to alter economic activity at such an alarming rate.

Scale, scope, and speed:

Digitalization has brought with it an abundance of products that have extremely small marginal costs (e.g. software). These products can scale very quickly, often with little labour (i.e. few employees). This development has also been accompanied by an increase in the complexity of ventures, incorporating numerous products, firms, or industries into a single project. This aspect is illustrated clearly by the classic iPhone example, which coordinates efforts between 43 countries and hundreds of firms (Apple, 2018). The management and logistics required to economically produce such a product would be virtually impossible without the innovations in communication technology brought by digitalization.

Ownership, assets and economic value:

In recent decades, the rise of intangible capital (e.g. software, data) has shifted investment interests. Knowledge-based capital has been receiving an increasing amount of business investment since the mid-2000s (OECD, 2013), and even outpaces investment in traditional capital in many countries (Corrado, Hulten, & Sichel, 2006). Digitalization has also brought about the emergence of numerous sources of value creation (e.g. sensors that will monitor asset performance and provide continual improvement through AI and data analysis). The nature of ownership has also been affected as the rise of digital platforms has greatly facilitated the monetization and sharing of physical capital.

Relationships, markets, and ecosystems:

Due to their intangible nature, things like software, data, and computing resources can be exploited from anywhere in the world, undermining the concept of value within geographical borders. With 58% of the global population now able to access the internet (Clement, 2019), it is feasible for the general population to contribute to the creation of their own networks. Not to mention the countless platforms and direct relationships that have been created as a result of digitalization, drastically reducing transaction costs in many markets.

After examining some of the mechanisms through which digitalization is transforming the global economy, it becomes clearer how it might affect various areas of policy. Among many others, the following are a few high-level policy examples with direct connections to the mechanisms we have considered (as discussed in Vectors of Digital Transformation):

- Changes to the rate at which firms can grow and acquire (or lose) market share suggest that policies maintaining low barriers to entry and innovation are important.
- Policies must reflect the fact that firms now have a much larger scope, and previously separate policy issue areas may now require coordinated efforts.
- A shift in investment incentives to better align with the digital economy may be of interest (e.g. R&D, data, IP).
- Policies hinged on geographical specifications (or defined markets) may be much less effective than in the past.
- Lastly, policies must consider the emergence of online platforms. The efficiency of markets has been altered, and the dynamics of competition (and maintaining a sufficient amount) have changed.

Digitalization must be taken into consideration when designing new policies or modifying old ones, and these mechanisms provide a basic understanding of how the digital transformation operates.

The economy is not what it was 30, 20 or even 10 years ago and policies must reflect this change to optimize their effectiveness. I find it fascinating how rapidly digitalization has altered the global economic landscape. In less than one generation, it has completely altered our economy, something that took the Industrial Revolution many times longer. Human society functions very differently today than it did 50 years ago, and 50 years from now too, it will operate entirely differently than it does today.

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